

**Y2K is back**

The millennium bug is making a resurgence because lazy devs postponed the problem instead of fixing it. <https://digit.in/apr20-21>

**Open source census**

The Linux foundation and LISH have published the second edition of the OS census. <https://digit.in/apr20-22>



Tracing the origins of the binary number system

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he formulation of the binary number system essentially laid the groundwork for digital circuitry,

computers, and the field of computer science, as we know it in today's technologically-advanced world. As our world has traversed technologically from simple mechanics all the way to quantum modeling, the need to count hasn't diminished over time, by humans and machines alike. The primary system used by humans for calculation is the Decimal Number System, however, the need for a more sophisticated, straightforward number system in digital computers and computer-based devices caused the adoption of the binary number system.

WHAT IS THE BINARY NUMERAL SYSTEM?

The binary number system is very literal to its nomenclature. Simply put, it is literally a numbering system that represents numbers using only two unique digits - typically, 0 and 1. The numbering system is also known as the base-2 number system. Computers utilise this numbering system to store and manipulate their data which includes numbers, words, music, graphics and more. In fact, the term 'bit', which is the smallest possible unit of digital technology, actually stems from the words 'BInary digiT'. Today, programmers use the hexadecimal or the base-16 number system as a more compact way to represent these binary numbers. Why? Because it is simpler for computers to convert from binary and hexadecimal and vice versa, and it is significantly harder to do this with the commonly-used decimal number system.

Are you still confused as to why the binary number system is used in computers? It's simple. No really, it is because of its simplicity. A computing system often hosts a series of switches

(electronic, mechanical, etc), each of which can alternate between the 'on' and 'off' positions. So, when a switch is on it is represented by the value of 1 and off is represented by 0. So, digital devices such as our computers perform mathematical operations using binary switches which are either on or off.

The concept of positional notation which is used in decimal numbers to determine the number using positional data is also applied in binary. However, in binary, the base number or radix is always 2. So, the binary number 1001 can be found in this manner - $1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 9$. Also, if you want to learn how to count in binary, here's a handy guide on Wikihow for the same (<https://digit.in/binary>).

THE EARLY ORIGINS BEFORE LEIBNIZ

The modern binary number system was first researched in Europe in the 16th and 17th centuries by Gottfried Leibniz, and a few other mathematicians. However, systems similar to the binary system appeared in the antiquity of various cultures and civilizations. The I Ching